

COLUMBIA GORGE, WA, USA

WMO: 726988

Lat: 45.619N

Lon: 121.166W

Elev: 72

StdP: 100.47

Time Zone: -8.00 (NAP)

Period: 00-19

WBAN: 24219

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-8.4	-5.9	-13.4	1.2	-6.7	-10.4	1.6	-4.4	11.0	7.9	9.0	8.2	1.4	110	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	36.8	19.2	34.6	18.8	32.4	18.2	20.8	32.8	19.9	31.5	19.1	30.2	4.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	11.7	25.3	15.4	11.0	24.7	14.5	10.4	24.0	60.0	33.0	56.7	31.5	54.0	30.3	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.7	10.7	9.3		-11.4	40.3	3.4	1.6	-13.9	41.5	-15.9	42.4	-17.8	43.3	-20.2	44.4
			DB	-12.1	22.3	3.2	0.8	-14.5	22.9	-16.3	23.3	-18.1	23.8	-20.5	24.3
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	2.8	4.6	8.2	11.7	16.4	19.8	24.1	23.7	19.2	12.5	6.0	2.3
	DBStd	8.39	3.94	3.71	3.31	3.06	3.58	3.55	3.10	2.96	3.23	3.45	4.17	3.84
	HDD10.0	851	225	154	73	16	1	0	0	0	14	129	239	
	HDD18.3	2564	483	385	315	201	82	23	1	1	27	183	368	497
	CDD10.0	1816	1	3	16	66	200	293	436	425	274	92	10	1
	CDD18.3	489	0	0	0	1	23	67	178	167	52	2	0	0
	CDH23.3	5780	0	0	1	44	348	783	2076	1856	640	32	0	0
	CDH26.7	2701	0	0	0	10	127	339	1059	914	246	6	0	0
Wind	WSAvg	3.5	1.8	2.2	3.1	4.2	4.7	5.4	5.7	4.9	3.7	2.5	1.9	1.8
Precipitation	PrecAvg	365	71	40	32	18	11	9	4	11	12	25	56	76
	PrecMax	499	192	108	81	39	32	25	19	48	34	71	148	230
	PrecMin	194	11	3	6	0	0	0	0	0	0	3	9	8
	PrecStd	83	44	27	21	12	9	8	5	13	10	20	32	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	16.0	21.8	28.2	33.5	37.4	39.7	39.4	34.9	27.0	18.2	13.2
		MCWB	9.8	10.8	12.6	15.1	17.7	19.2	19.9	19.3	18.6	15.6	13.1	10.0
	2%	DB	11.5	13.7	18.3	23.8	30.1	33.6	37.0	36.5	32.0	23.3	15.7	10.8
		MCWB	8.3	9.3	10.7	13.1	16.6	18.6	19.4	18.9	17.3	14.2	11.7	7.9
	5%	DB	9.6	12.0	16.2	20.8	27.2	30.5	34.9	34.2	29.7	20.8	13.8	9.0
		MCWB	7.0	7.7	9.9	11.9	15.5	17.1	19.1	18.7	16.8	13.0	9.9	6.6
	10%	DB	7.9	10.2	14.1	18.3	24.4	27.8	32.4	31.8	27.2	18.8	12.0	7.4
		MCWB	5.9	6.6	9.0	10.8	14.4	16.4	18.6	18.1	15.8	12.4	9.0	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	11.5	13.6	15.7	19.0	21.1	21.9	21.8	20.0	17.1	14.6	10.7
		MCDB	13.4	15.0	20.0	25.8	30.3	33.5	34.5	33.6	30.9	24.0	17.4	12.3
	2%	WB	8.7	9.7	11.7	14.0	17.5	19.3	20.9	20.6	18.7	15.4	12.1	8.3
		MCDB	11.1	13.4	16.5	21.7	28.6	31.0	33.5	31.9	29.0	21.0	14.7	10.3
	5%	WB	7.4	8.1	10.5	12.6	16.2	18.1	20.0	19.7	17.7	14.2	10.6	6.9
		MCDB	9.4	11.3	14.9	19.9	25.4	29.2	32.1	31.0	27.2	19.4	13.2	8.8
	10%	WB	6.1	7.0	9.4	11.3	15.1	16.9	19.1	18.9	16.7	13.2	9.3	5.5
		MCDB	7.9	9.7	13.7	17.3	23.0	26.4	30.7	29.7	25.5	17.9	11.6	7.2
Mean Daily Temperature Range	5% DB	MDBR	5.6	8.6	10.7	11.8	13.0	12.7	14.6	14.6	14.5	11.8	7.9	5.4
		MCDBR	8.9	11.5	14.4	17.5	18.9	18.5	19.2	19.5	19.5	15.4	10.1	8.2
		MCWBR	6.1	7.2	8.2	8.5	7.7	6.7	6.7	6.6	7.9	7.7	6.4	5.7
	5% WB	MCDBR	8.5	9.9	12.5	15.2	16.0	16.3	16.7	16.2	16.4	12.6	9.1	7.9
		MCWBR	6.0	6.5	7.4	7.6	6.8	6.3	6.1	5.8	6.8	6.9	6.0	5.6
Clear-Sky Solar Irradiance	taub	0.328	0.325	0.338	0.374	0.387	0.374	0.368	0.377	0.361	0.352	0.337	0.325	
	taud	2.417	2.479	2.461	2.346	2.336	2.392	2.427	2.409	2.431	2.441	2.423	2.403	
	Ebn at Noon	771	858	895	885	881	892	893	872	856	809	751	729	
	Edh at Noon	73	82	96	117	122	116	111	109	98	84	72	67	
All-Sky Solar Radiation	RadAvg	1.23	2.23	3.48	5.10	6.26	7.11	7.69	6.49	4.83	2.91	1.57	1.01	
	RadStd	0.16	0.31	0.27	0.36	0.42	0.34	0.31	0.34	0.26	0.26	0.15	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	+44

Nomenclature: See separate page